SIMATS ENGINEERING
ASSESSMENT TOOL 2
Case study

COURSE NAME: Computer Networks **COURSE CODE: CSA07**

COURSE OUTCOME COVERED

CO3: *Analyze the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions*

Assessment Tool Weightage: 40%

Code of Conduct:

I V.Sanjai(192312486) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

V-Sanjai
Signature of the student:

20/20

Cloud-Based Video conferencing:

1. Transport protocol or the application layer.

The issue is caused mainly by network condition and application layer buffering, not by the transport protocol ~~itself~~

→ The reported 8% packet loss 180 ms jitter, and fluctuating bandwidth indicate unstable network performance.

→ The application's buffering strategy may also be insufficient to compensate for high jitter.

2. Justification:

Real time multimedia application require.

→ Low latency (less than 150 ms preferred).

→ Low jitter (less than 30 ms).

→ Minimal packet loss (below 1%).

UDP is preferred because:

→ No retransmission.

→ Faster delivery.

→ Supports real-time voice and video.

3. Recommendations.

→ continue using UDP.

→ Enable RTP (Real time Transport Protocol).

→ Use RTCP for network monitoring.

Application-level Improvements:

→ Use adaptive jitter buffers.

→ Apply adaptive bitrate streaming.

→ Enable Forward Error correction.

→ Increase bandwidth during peak hours.

(5) DNS Infrastructure Optimization.

DNS resolution time.

1. Reasons for high DNS lookup time (300 ms) may occur because of:
   - → Centralized DNS server.
   - → DNS server overload.
   - → Cache misses.
   - → High network congestion.
   - → Poor TTL configuration.

2. Proposed DNS Architecture.

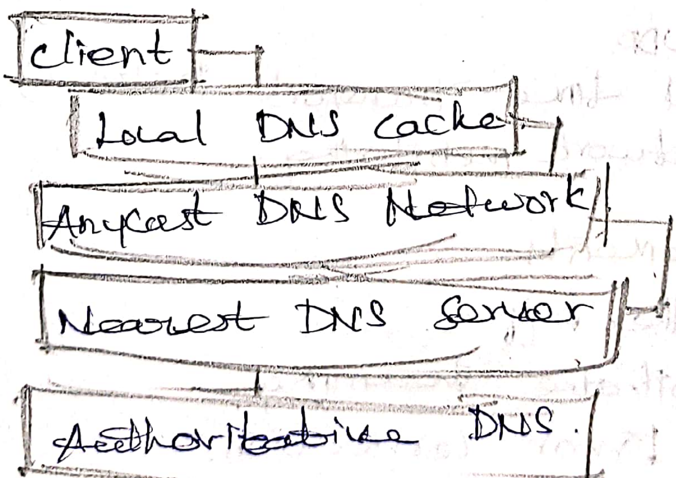
Use:

→ Anycast DNS.
(a) Multiple DNS servers share the same IP.

→ DNS Pre-fetching.
(b) Frequently accessed domains are resolved before user request them.

→ TTL optimization.
(a) Static records: TTL = 1-24 hours.
(b) Dynamic records: TTL = 60 - 300 seconds.

Architecture:



Advantages and Limitation:

Advantages:
- → Faster DNS resolution.
- → Better scalability.
- → High availability.
- → Automatic failover.
- → Lower server load.

Limitations:
- → Higher deployment cost.
- → More complex configuration
- → Cache inconsistency with very high TTL.
- → Requires global Anycast support.

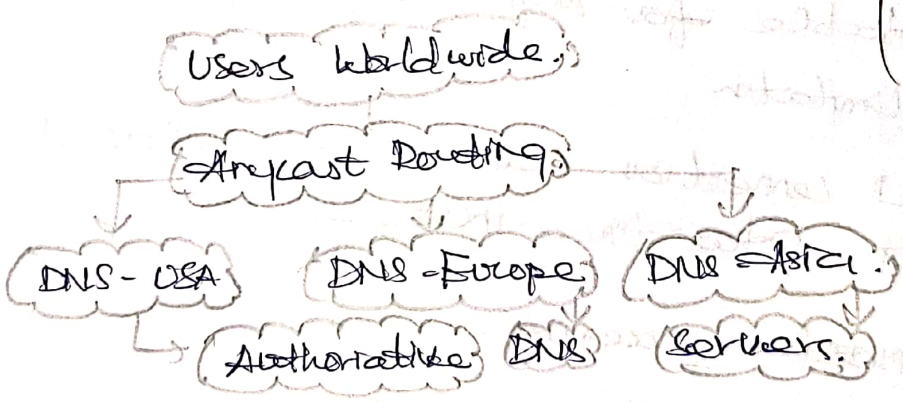
③ Distributed DNS for cloud services:

4. Impact of centralized DNS.

A centralized DNS architecture causes
- → High lookup latency.
- → Single point of failure.
- → Server overload.

Result:
- → DNS delay increases to around 800 ms.

5. Distributed DNS solution.



Features:
- → Multiple DNS servers
- → Anycast routing.
- → DNS caching.

6. Influence of TTL, Anycast, and DNS caching.

High TTL.
→ Less DNS traffic.
→ Slower updates.

Low TTL.
→ Faster record updates.
→ More DNS queries.

DNS caching.
→ Stores previous DNS responses.
→ Reduces lookup time.

(4) TCP performance in stock Trading Platform.

1. Three TCP-Related factors.

a) Flow control.
→ Small receive window limits data transmission
→ Causes delays.

b) Nagle's Algorithm.
→ Introduces additional delay.
→ Unsuitable for high-frequency trading.

c) SYN queue limitation.
→ New TCP connection wait or are dropped.
→ Connection setup time increase.

2. Effect on transaction processing.

Factor	Effect.
Flow control.	Slower order processing due to reduced transmission rate.
Nagle's Algorithm	Delays sending small buy/sell orders.

TCP configuration Recommendation.

→ Disable Nagle's Algorithm.

→ Increase TCP receive / send buffer size.

→ Enable TCP Fast Open.

→ Increase SYN backlog queue

→ Reduce excessive buffering.

→ Enable Selective Acknowledgment (SACK).